Wizard Quest

Sprint 2 presentation

Team members:

- Emre Acarsoy
- Luca Croci
- Tom Croft
- Will Finney
- Kazybek Khairulla
- Luca Pacitti
- Harry Taylor

Structure

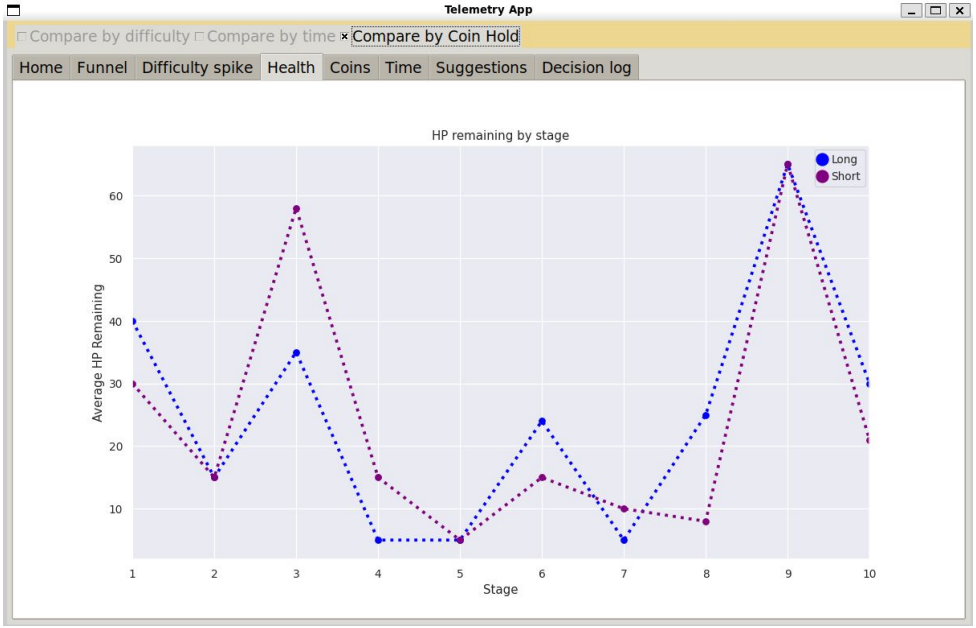
1) Problem statement
2) Improvements from Sprint 1
   a) -> Game
   b) -> Telemetry
3) Live demo
4) Evaluation
5) Ethical & Legal considerations
6) Scrum, and best practices

Sprint 2 Roles

Emre Acarsoy	Scrum master & Python developer
Luca Croci	UI & UX Lead & Java developer
Tom Croft	Game Lead
Will Finney	Java developer & Project designer
Kazybek Khairulla	Java developer
Luca Pacitti	Testing Lead & Java developer
Harry Taylor	Telemetry Lead & Python developer

Problem Statement

- Game balancing is difficult for designers.
- They often have to rely on their own intuition and small-scale playtests.
- This makes it difficult to identify problem areas where players lose engagement.
- What if we had live analytics generated whilst players engage?



Improvements from Sprint 1

Game

- Command line ⟶ GUI
- 2 encounters ⟶ 10 encounters
- 1 adjustable game parameter ⟶ 8 adjustable game parameters
- Time-realistic simulations

Telemetry

- Fairness view
- Comparison modes
- 1 dynamic suggestion ⟶ 8 dynamic suggestions
- Export to csv functionality
- Decision log

Improvements from Sprint 1

Main landing page

```
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 64 source files with javac [debug release 17] to target/classes
[INFO]
[INFO] --- exec-maven-plugin:3.3.0:java (default-cli) @ wizard-quest ---
 _    _   _                            _   _____                        _
| |  | | (_)                         | | |  _  |                      | |
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Signing in via Google...
Welcome, Emre Acarsoy! (Role: DEVELOPER)

Main Menu
1. Start New Game
2. Settings
3. View Telemetry Disclosure
0. Quit
>>>
```
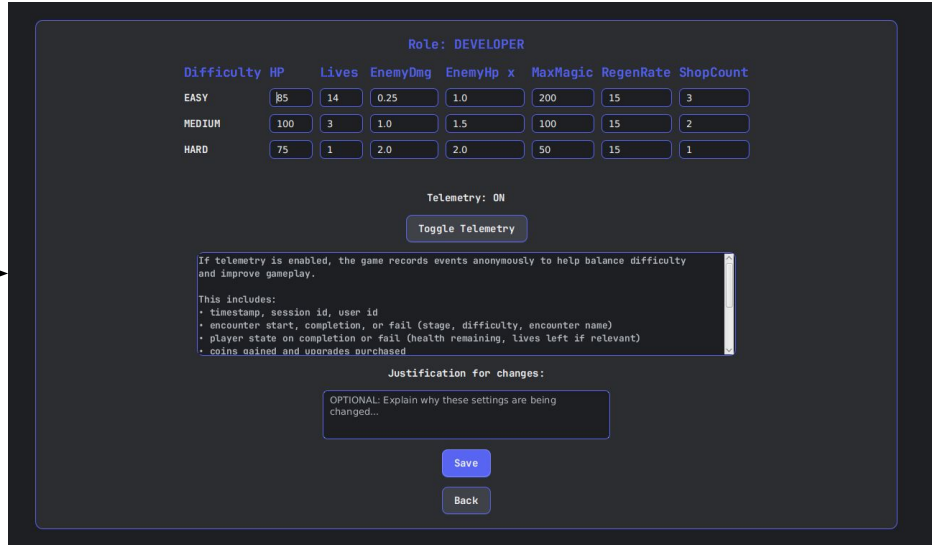

Improvements from Sprint 1

Settings page

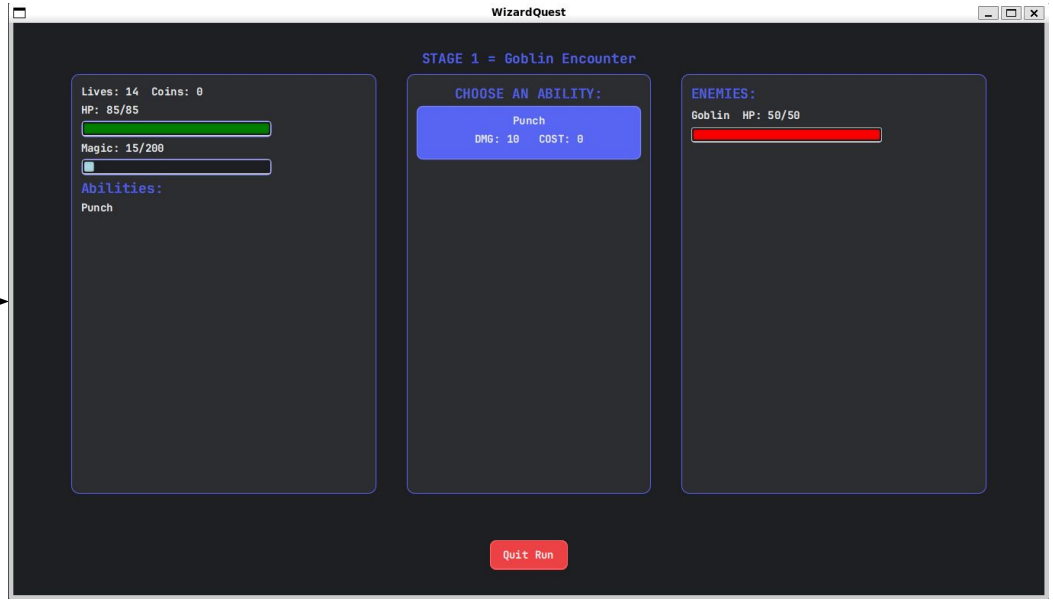
```
Settings
User role: DEVELOPER
Telemetry: ON

1. Toggle telemetry
2. Telemetry disclosure
3. Change starting lives (design parameter)
4. Run Simulated Game
0. Go Back
>>>
```



Improvements from Sprint 1

Encounters

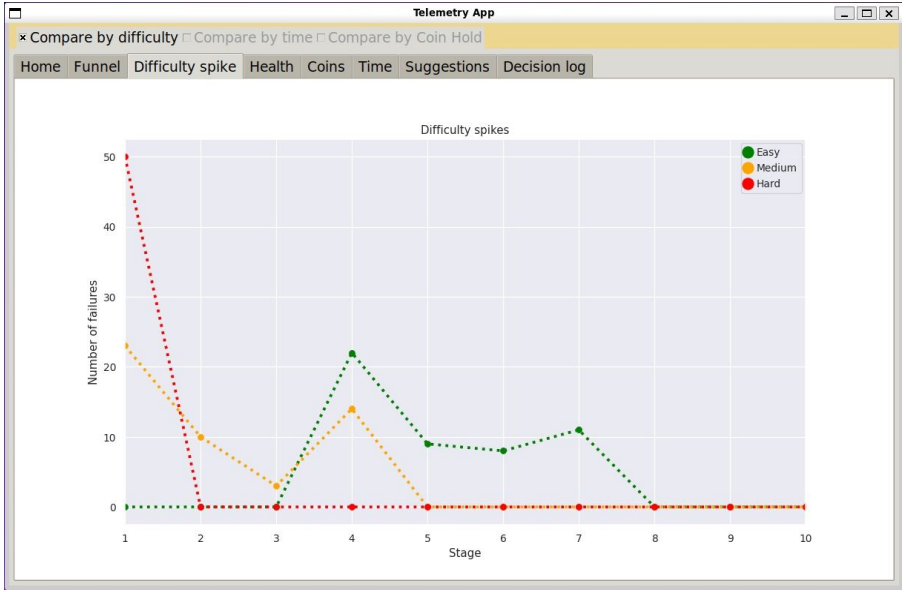
```
Main Menu
1. Start New Game
2. Settings
3. View Telemetry Disclosure
0. Quit
>>> 1

Select Difficulty
1. Easy [Starting Lives: 10]
2. Medium [Starting Lives: 3]
3. Hard [Starting Lives: 1]
0. Go Back
>>> 1

Stage 1
Encounter: Fish Man Encounter
        Fish Man (Health: 50/50)

Health: 200 / 200
Magic: 15 / 200
Lives: 10
Coins: 0

Choose an ability:
1. Punch (10 damage) - A simple punch. (Cost: 0 magic)
0. Quit Run
>>> |
```



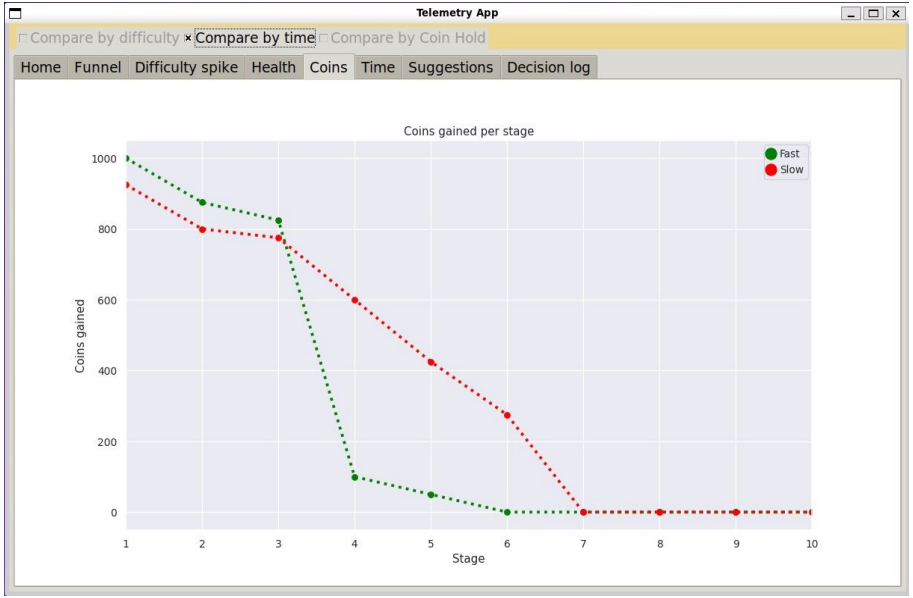
Improvements from Sprint 1

Telemetry improvements

Difficulty comparisons

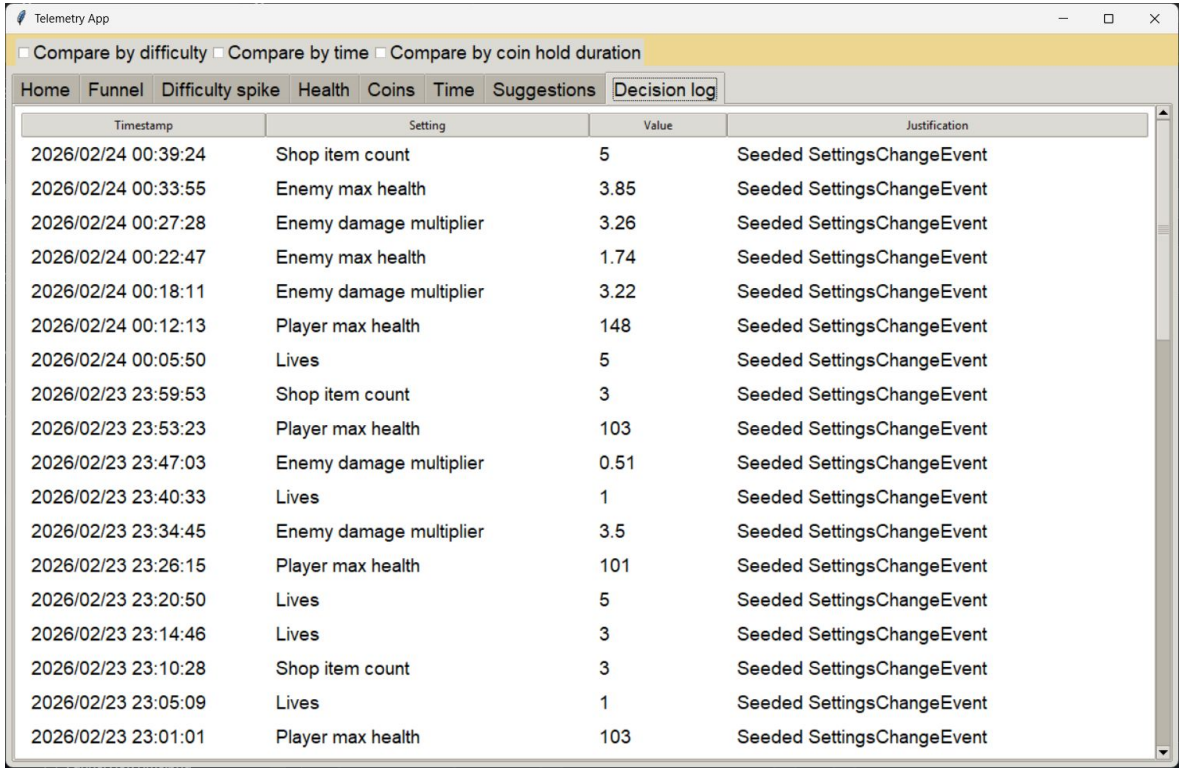


Speed comparisons



Improvements from Sprint 1

Decision log



Timestamp	Setting	Value	Justification
2026/02/24 00:39:24	Shop item count	5	Seeded SettingsChangeEvent
2026/02/24 00:33:55	Enemy max health	3.85	Seeded SettingsChangeEvent
2026/02/24 00:27:28	Enemy damage multiplier	3.26	Seeded SettingsChangeEvent
2026/02/24 00:22:47	Enemy max health	1.74	Seeded SettingsChangeEvent
2026/02/24 00:18:11	Enemy damage multiplier	3.22	Seeded SettingsChangeEvent
2026/02/24 00:12:13	Player max health	148	Seeded SettingsChangeEvent
2026/02/24 00:05:50	Lives	5	Seeded SettingsChangeEvent
2026/02/23 23:59:53	Shop item count	3	Seeded SettingsChangeEvent
2026/02/23 23:53:23	Player max health	103	Seeded SettingsChangeEvent
2026/02/23 23:47:03	Enemy damage multiplier	0.51	Seeded SettingsChangeEvent
2026/02/23 23:40:33	Lives	1	Seeded SettingsChangeEvent
2026/02/23 23:34:45	Enemy damage multiplier	3.5	Seeded SettingsChangeEvent
2026/02/23 23:26:15	Player max health	101	Seeded SettingsChangeEvent
2026/02/23 23:20:50	Lives	5	Seeded SettingsChangeEvent
2026/02/23 23:14:46	Lives	3	Seeded SettingsChangeEvent
2026/02/23 23:10:28	Shop item count	3	Seeded SettingsChangeEvent
2026/02/23 23:05:09	Lives	1	Seeded SettingsChangeEvent
2026/02/23 23:01:01	Player max health	103	Seeded SettingsChangeEvent

Demo!

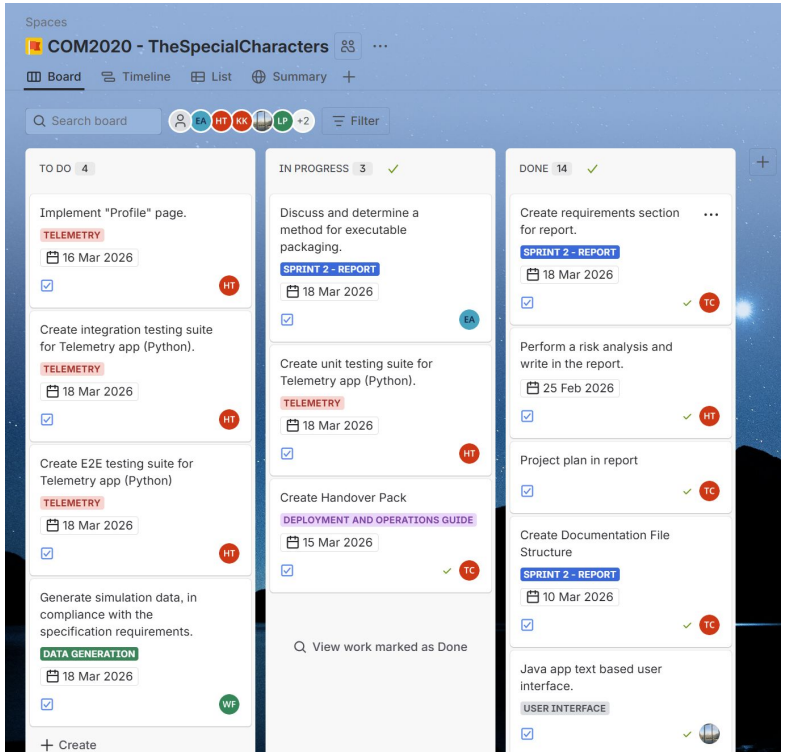
Evaluation

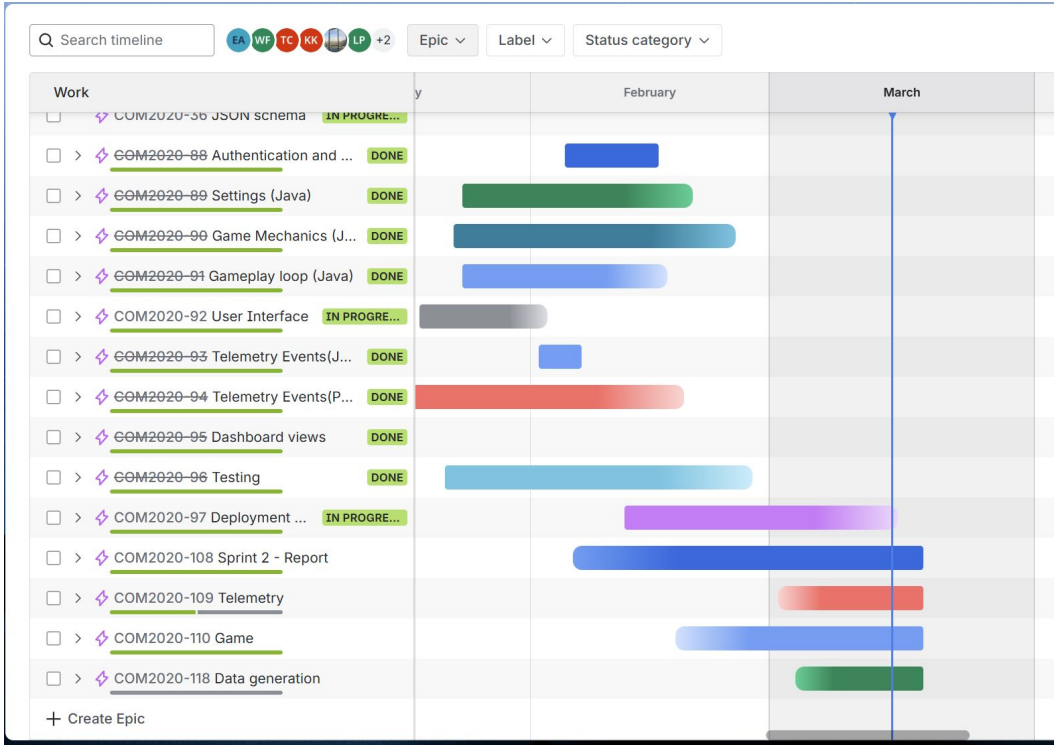
- Telemetry system records player actions, encounters, and outcomes
- Game data can be used to identify balancing issues and difficulty spikes
- Telemetry runs in the background and does not interrupt gameplay
- Provides objective data to support balancing decisions and suggestions

Ethical and Legal considerations

- **Privacy and Data Protection**
  - Google OAuth | Data is pseudonymised
- **Consent and Data Disclosure**
  - Balancing purposes | Can be disabled
- **Accessibility**
  - High-contrast. Keyboard navigation.
- **Intellectual Property and Licence**
  - Copyright. MIT Licence.

Scrum Board: Jira tasks, stories, epics





Continuous integration:

All checks have passed
2 successful checks

No conflicts with base branch
Merging can be performed automatically.

Pull requests:

2 Open ✓ 261 Closed

- Fairness indicators implemented feature
  #321 by ht555-EXE was merged 2 weeks ago
- Updated suggestion parameters improvement
  #320 by ht555-EXE was merged 2 weeks ago
- Add justification field to Telemetry GUI improvement
  #319 by AtlasICL was merged 2 weeks ago
- Add justification field to SettingsChange events improvement
  #317 by AtlasICL was merged 2 weeks ago
- Meeting minutes for 03-11 documentation
  #316 by lucaapacitti was merged 2 weeks ago
- Update today's meeting notes before meeting documentation
  #315 by AtlasICL was merged 2 weeks ago
- Added functionality for resetting all file sources. bug improvement
  #312 by AtlasICL was merged 2 weeks ago
- Convert data source selection to dropdown menu improvement
  #310 by AtlasICL was merged 2 weeks ago
- Small improvements for suggestions text bug codestyle
  #305 by AtlasICL was merged 2 weeks ago
- Method name bug fix bug
  #303 by ht555-EXE was merged 2 weeks ago
- Implement decision log view (telemetry app) feature
  #302 by AtlasICL was merged 2 weeks ago

Issues:

Open 5 Closed 142 Author

- Telemetry not being recorded in java bug
  #279 · by Tom-Croft was closed 2 weeks ago
- More items in shop feature question
  #278 · by Tom-Croft was closed last week
- GUI improve environment variable not set error improvement user interface
  #277 · by Tom-Croft was closed 2 weeks ago
- GUI shows incorrect stage reached bug user interface
  #276 · by Tom-Croft was closed 2 weeks ago
- GUI magic incorrectly handled bug user interface
  #275 · by Tom-Croft was closed 2 weeks ago
- "Quit Game" button in game GUI does not send back to main menu bug
  #271 · by AtlasICL was closed 2 weeks ago
- Should all views in telemetry be per-difficulty? question
  #265 · by AtlasICL was closed 2 weeks ago
- Implement "Fairness indicators" feature
  #264 · by AtlasICL was closed 2 weeks ago
- Implement telemetry page for time-to-complete feature
  #263 · by AtlasICL was closed 2 weeks ago
- Complete python side encounter enums. bug documentation
  #258 · by ht555-EXE was closed 2 weeks ago
- Missing testing instructions in README documentation
  #257 · by AtlasICL was closed 2 weeks ago

Thank you!